



Municipal Bond Pricing Project

Georgia Institute of Technology: CSE 6242: Data & Visual Analytics

Team #153: 2Legit2Quit - Jacob Burbank-Goldrich, Janarthanan Gopi, Issac J Kuo, Saswata Misra, Constance Julia Nething, and Angela Tran



Background

Municipal (muni) bonds are a debt security used by local governments to fund public projects such as schools, roads, airports, etc. These investments are attached to tax implications and other many factors making this type of investment complex and pricing inefficient.

Objective & Impact

To provide traders with a tool to find similar bonds in order to price a prospective bond accurately.

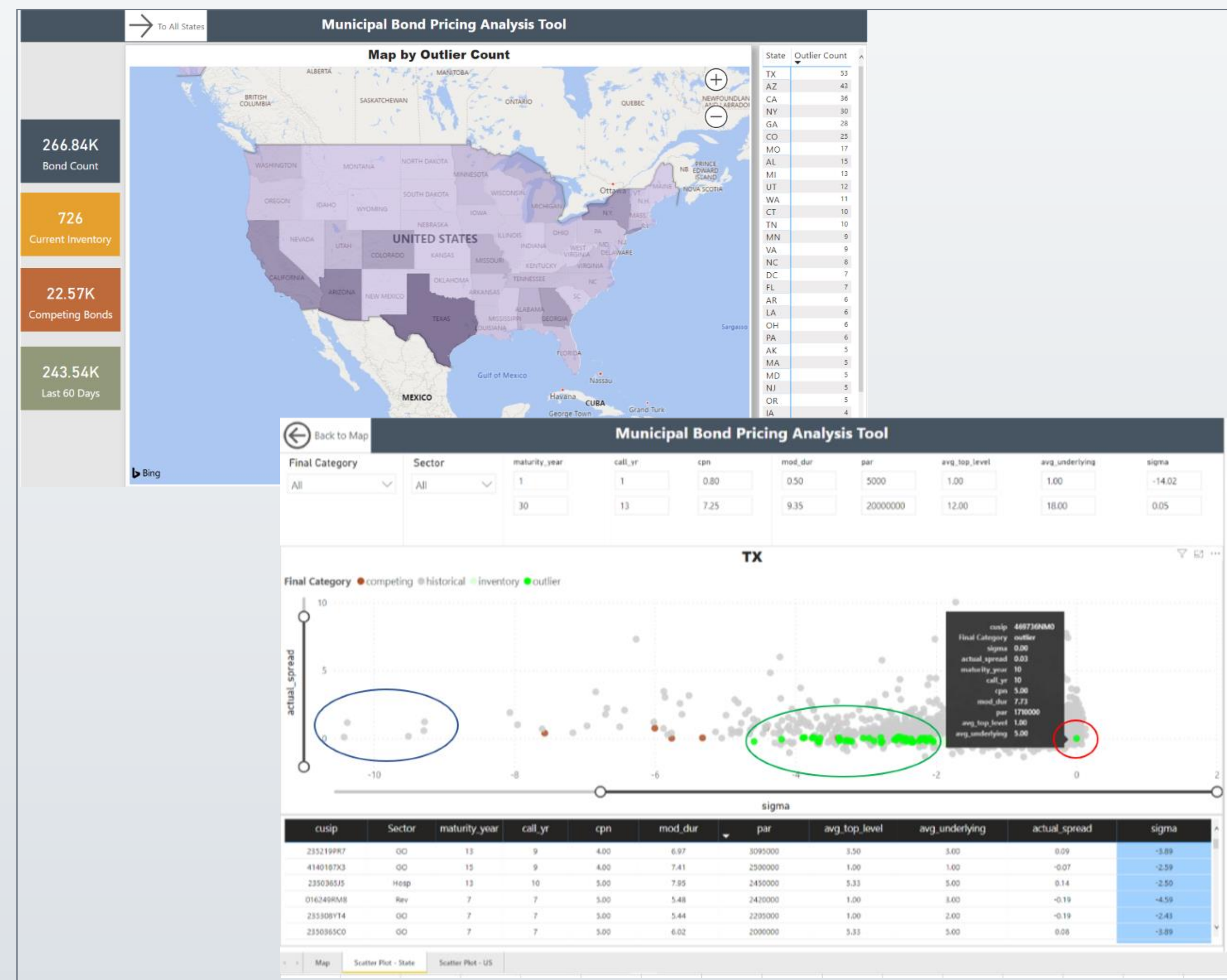
Pricing of US municipal bonds is both opaque and inefficient, owing to high levels of complexity and the differing information available to interested parties. These pricing inefficiencies can result in various municipalities being unable to initiate or fund projects due to being unable to match issuer and investor. In addition, information asymmetry (unequal information between parties) contributes to pricing inefficiency.

The objective of this project is to use machine learning to group similar bonds in order to help traders optimally price their inventory by identifying outliers and gain insight into changing market conditions. Successful deployment of this solution will allow traders price bonds appropriately and increase profit. A secondary objective is to allow traders to visualize outliers in traded bonds (historical) to identify new investment opportunities.

Methodology

Previous solutions have not incorporated a configurable visual element to aide in further refining their analysis. We are proposing a solution that will combine a configurable visual component with traditional machine learning models.

This new approach will be successful because it will allow traders to visualize the pricing of their bonds against comparable bonds, which also reduces required research time. As a result, this will also increase transparency and efficiency in the market.



Acquire Data across 3 Datasets

Historical bond trades over the past 60 days downloaded from the Municipal Securities Rule Making Board (MSRB). Competing bonds available for sale downloaded from Bloomberg Runz.

Current inventory of bonds was provided by team member's firm. Over 260,000 records with 17 quantitative and 6 categorical attributes

Analyze

Trained an Adaboost Decision Tree regressor and used it to predict the spreads/price on our inventory of bonds.

Calculated the difference between the model predicted spread and the actual spread to calculate an outlier score.

Visualize

Designed interactive visualizations to allow user to effectively explore the results and identify outliers.

Experiment and Evaluation

Clustering Algorithms:

- Gower Distance & K-medoids
- K-Means
- PCA + DBSCAN
- Kamila
- FAMD Clustering

Regression:

- Multivariate linear regression
- Box Cox transformation
- Interaction terms
- Polynomial terms
- Ridge Regression
- Lasso
- Adaboost Decision Tree Regressor

We used the "eval_spread" to verify the quality of the clustering algorithms. In an ideal cluster of similar bonds, the eval_spread should be very close in value. More specifically for each cluster we calculated the standard deviation of the eval_spread of each bond vs the median eval_spread. We then calculated the weighted average standard deviation of eval spread across all clusters ("C-Score"). The best result from any of the clustering algorithms produced a weighted average standard deviation of 0.2874 which we deemed to be too large.

The final regression model was the Adaboost Decision Tree regressor with the Box Cox transformation, interaction terms and polynomial terms. That produced a root mean squared error of 0.1546. We used this model together with the visualization to effectively narrow down the population of over 260,000 bonds and identify only the handful of pricing outliers. The user can then use the tool to drill down and not only correct pricing on their inventory but look at the outliers of historical trades to identify opportunities.